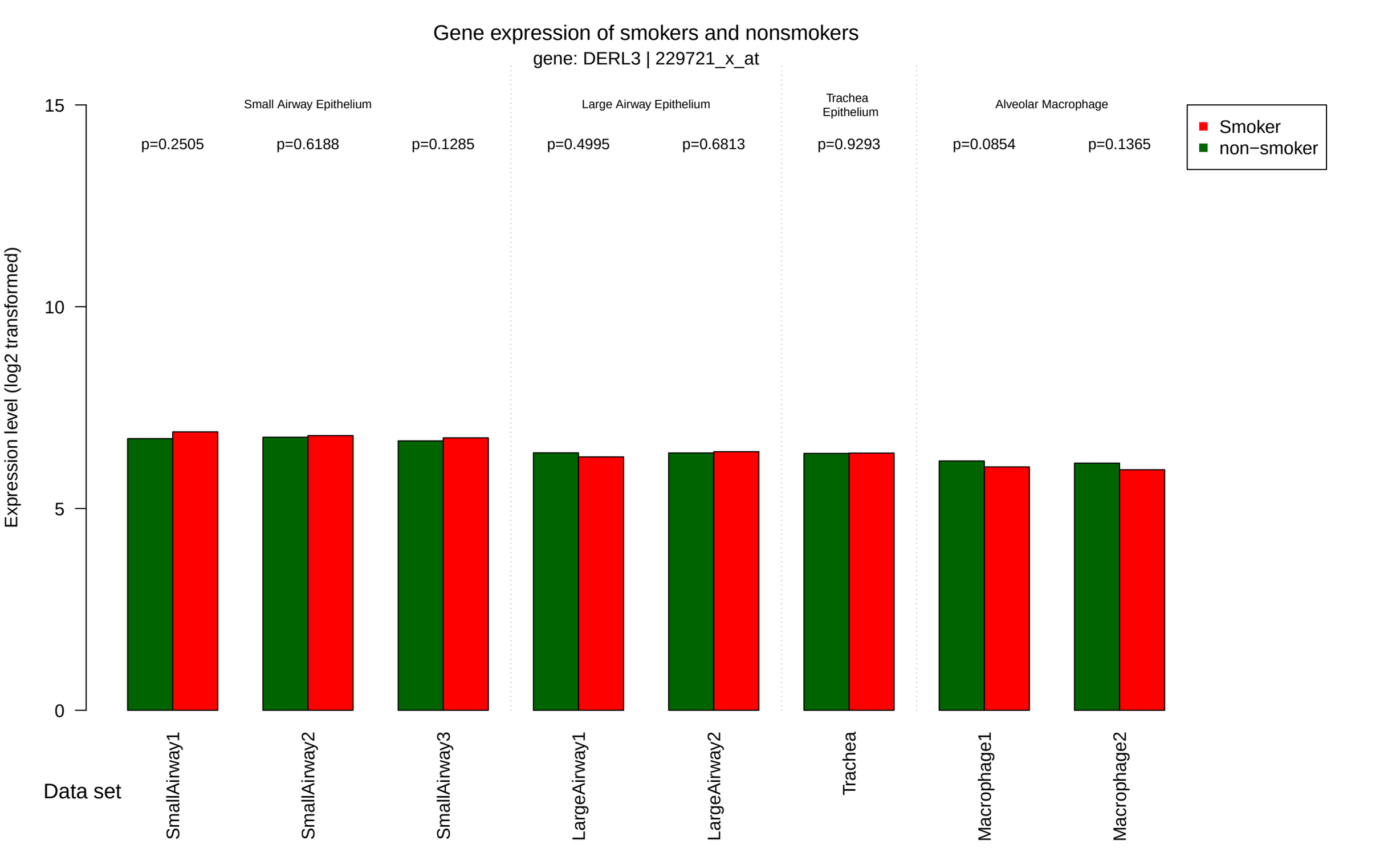
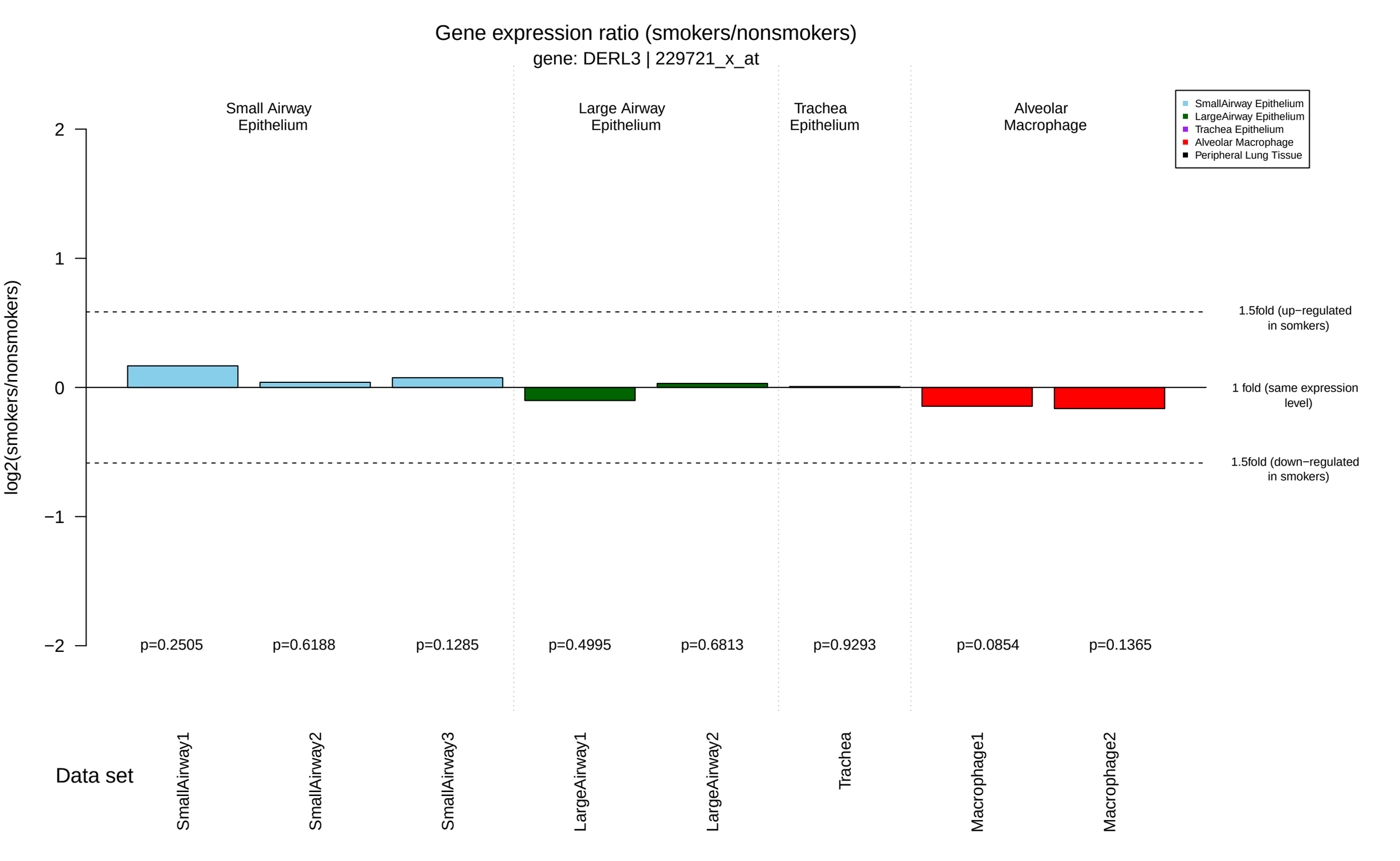
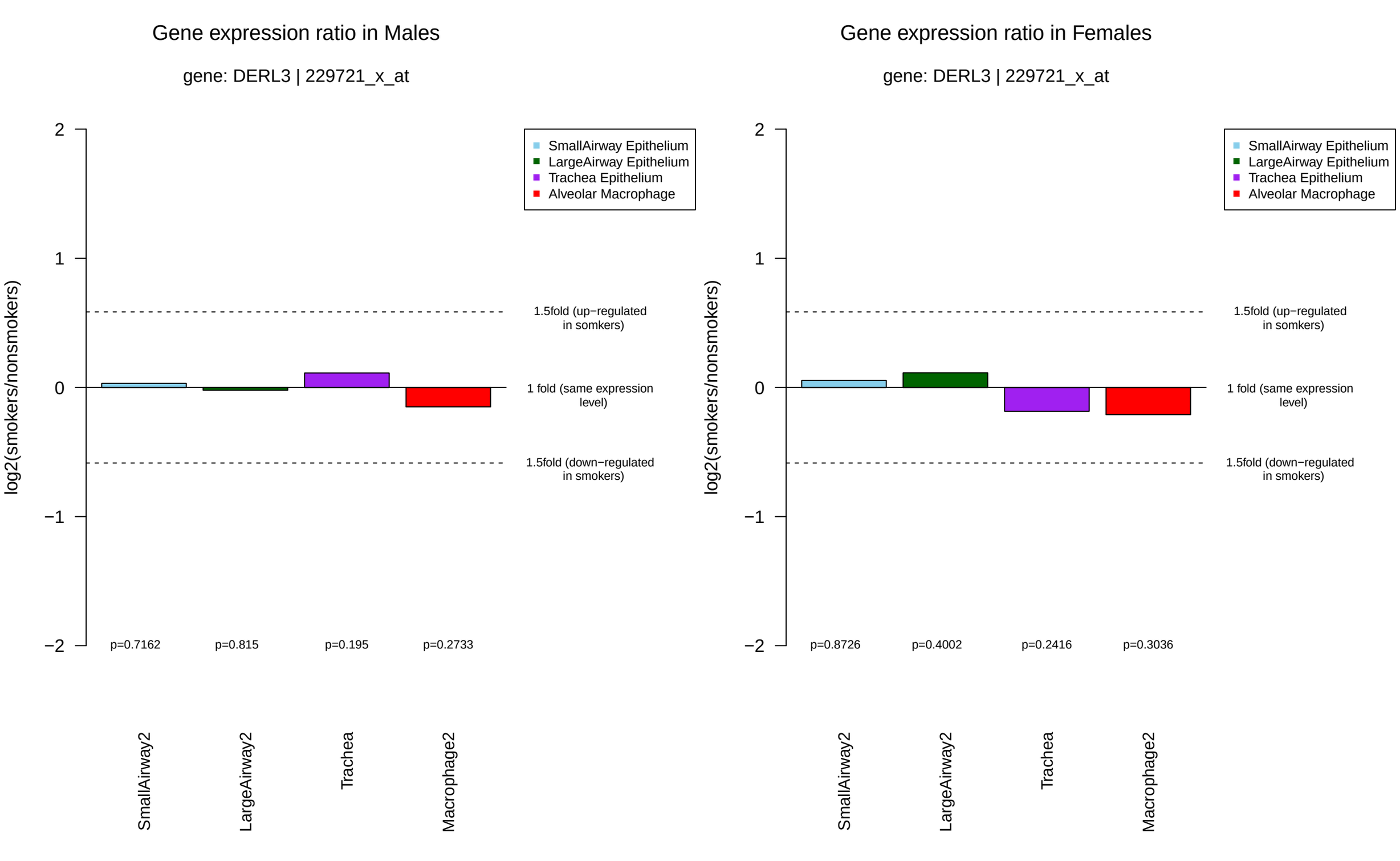
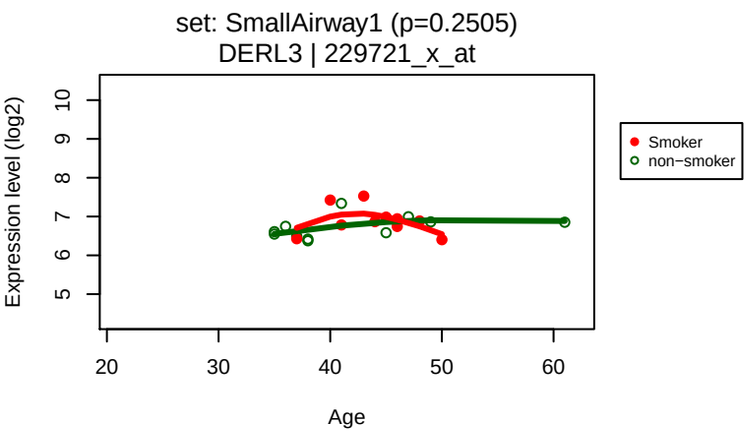
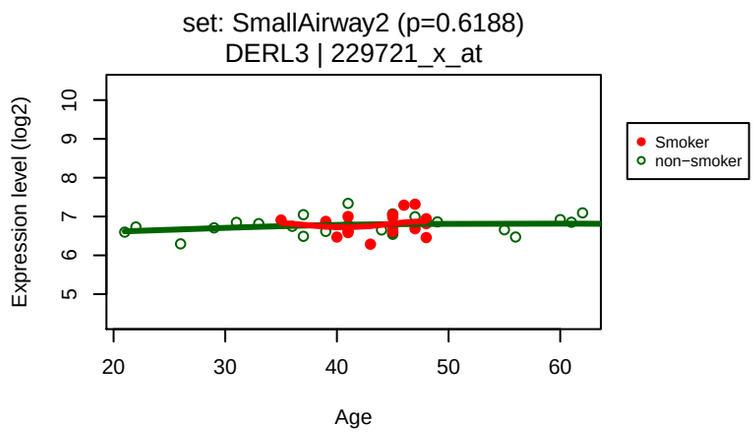
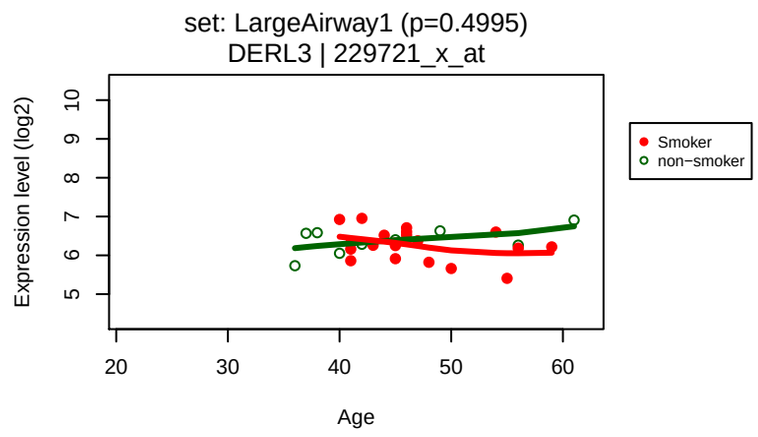

Gene expression of smokers and nonsmokers
gene: DERL3 | 229721_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2505
p=0.6188
p=0.1285
p=0.4995
p=0.6813
p=0.9293
p=0.0854
p=0.1365
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

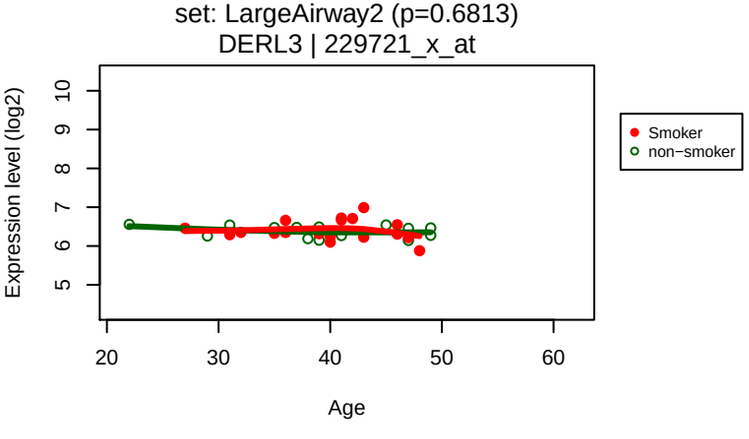

Gene expression ratio (smokers/nonsmokers)
gene: DERL3 | 229721_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2505
p=0.6188
p=0.1285
p=0.4995
p=0.6813
p=0.9293
p=0.0854
p=0.1365
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

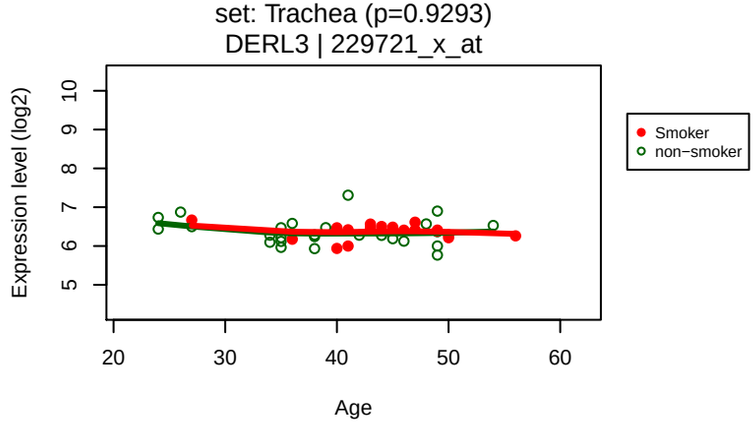

Gene expression ratio in Males
gene: DERL3 | 229721_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7162
p=0.815
p=0.195
p=0.2733
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DERL3 | 229721_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8726
p=0.4002
p=0.2416
p=0.3036
SmallAirway2
LargeAirway2
Trachea
Macrophage2

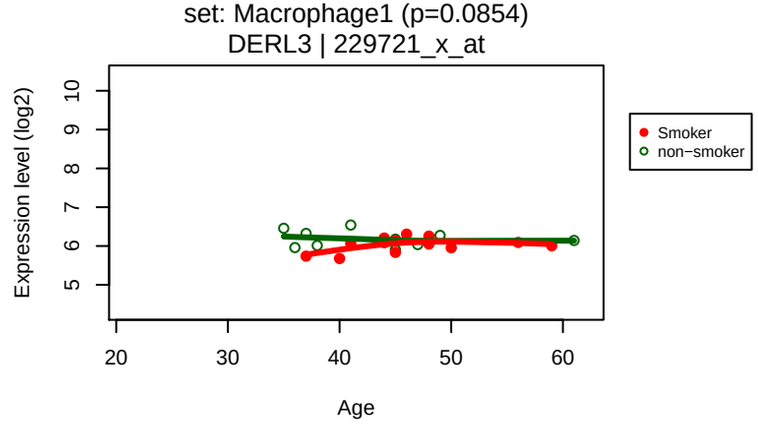

set: SmallAirway1 (p=0.2505)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker

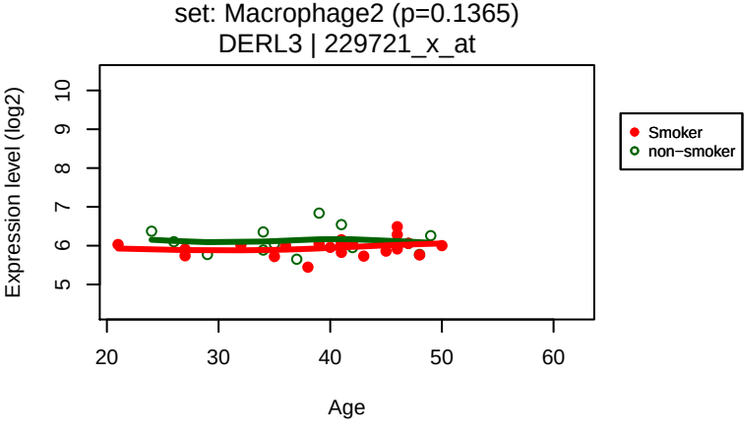

set: SmallAirway2 (p=0.6188)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker

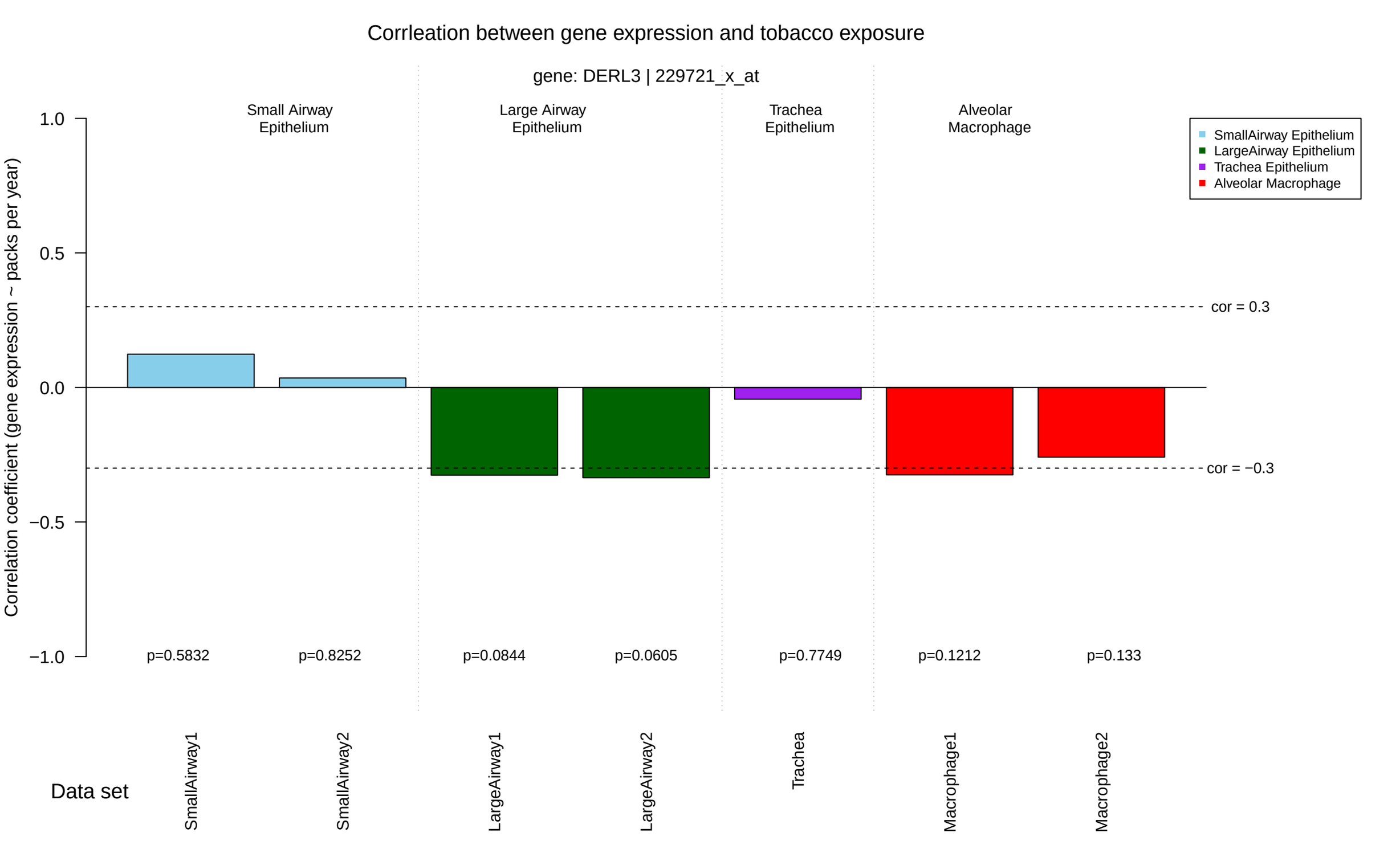

set: LargeAirway1 (p=0.4995)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6813)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9293)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0854)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1365)
DERL3 | 229721_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DERL3 | 229721_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5832
p=0.8252
p=0.0844
p=0.0605
p=0.7749
p=0.1212
p=0.133
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2